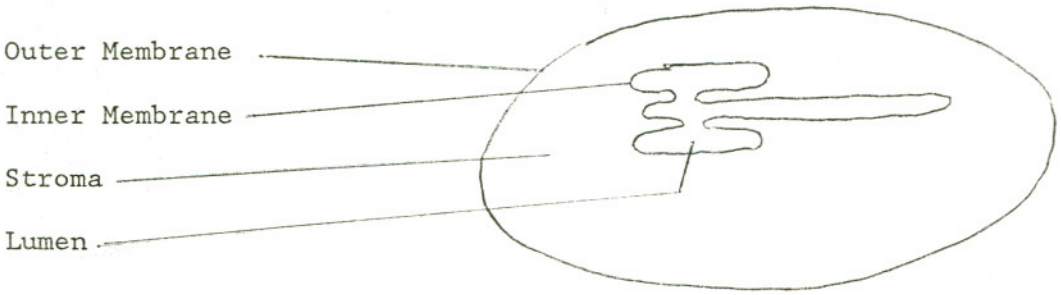
BIOCHEMISTRY CUMULATIVE EXAMINATION

March 31, 1990

Plastocyanin is a small protein of about 10 kd which is located in the lumen (inside the inner membrane) of the chloroplast. The gene for plastocyanin is located on nuclear DNA and indicates the protein to be synthesized in a precursor form which is approximately 15 kd in size. The protein is synthesized on cytoplasmic ribosomes.



The gene for plastocyanin is described below.

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                                Portion of gene describing sequence
                                 for 10 kd mature form of protein
|-------------------------|-----------------------------------|
A                         A
 B          B
             C           C
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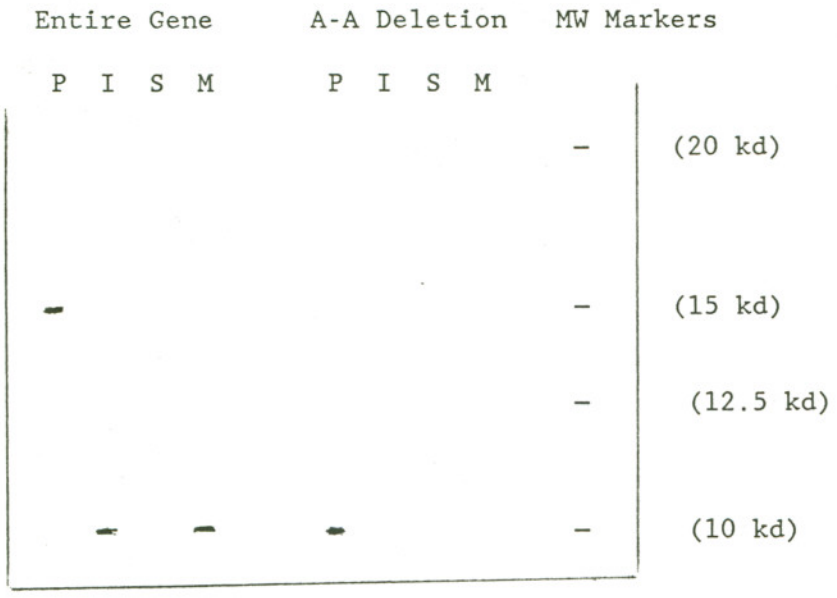
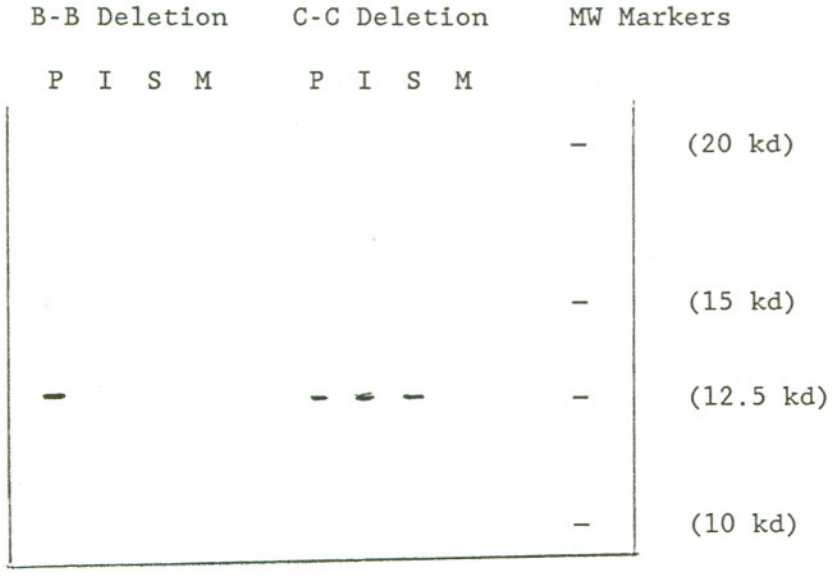
A, B, and C indicate cleavage sites for restriction enzymes.

To examine the role of the extra 5 kd of protein present in the precursor form, the plastocyanin gene has been cloned and the restriction enzyme cleavage sites in this region used to generate altered genes in which some or all of this region is missing. The resulting genes are transcribed and translated using in vitro methods in which the protein product is radiolabelled.

The protein product obtained is examined as to whether it is taken up and processed by isolated intact chloroplasts. The procedure involves adding the protein product to intact chloroplasts and incubating in the presence of light for the desired period of time. At the end of the incubation period, the presence or absence of plastocyanin is examined by SDS gel electrophoresis and autoradiography. Three situations are examined: (I) intact chloroplasts (i.e., with outer membrane intact), (S) stromal fractions generated by rupture of the outer membrane of the intact chloroplasts, and (M) the intact inner membrane fraction isolated from intact chloroplasts. (P) indicates the protein product of the in vitro translation system in each case which was also examined by SDS gel electrophoresis prior without exposure to chloroplasts.





a. (40 points) Based on the results shown above, what would you conclude about the role(s) of the extra 5 kd of protein in the plastocyanin precurcor? Be as specific as possible and explain the basis for your conclusions.

b. (30 points) If the same experiment were done in the absence of light, no uptake or processing of the plastocyanin precursor was observed. What might be a likely explanation for this? How might you test your hypothesis?

c. (30 points) Let us assume that you wish to repeat the experiment but instead of using intact chloroplasts, you use a preparation of inner chloroplast membranes which is lacking the outer membrane and stroma contents. How would the results differ? Explain your answer in terms of your conclusions in part a.